[AD] — Analytical Development

Drift, Diffusivity, the Escape Law, the Convention, and the MFPT Quadrature

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1 Setup, notation, and standing assumptions

1.1 The model

Position evolves by the overdamped discrete-time update

$$x_{t+1} \;=\; x_t + v\, a_t\, \Delta t + \sqrt{2D\Delta t}\, \xi_t, \qquad \xi_t \sim \mathcal{N}(0,1) \text{ i.i.d.}, \tag{1}$$

with action $a_t \in \{-1, +1\}$, swimming speed v, and motility diffusivity D. The agent never observes x_t. It observes

$$\tilde{c}_t \;=\; c(x_t) + \sigma \eta_t, \qquad \eta_t \sim \mathcal{N}(0,1) \text{ i.i.d.}, \tag{2}$$

where c is the double-Gaussian landscape of Week 1 and σ is the sensing-noise strength. The two noises are physically distinct and must never be merged: σ corrupts *information*, D corrupts *motion*.

1.2 Standing assumptions

Three assumptions are used throughout and should be checked, not assumed, once code exists.

(A1) **Smooth landscape on the step scale:** $v\Delta t \ll w_1$, so that $c'(x)$ may be treated as constant across one step and the diffusion approximation to the discrete chain is valid.

(A2) **Separation of timescales:** the internal memory state equilibrates fast compared to the barrier-crossing time, so that a Markovian Fokker–Planck description in x alone is legitimate after coarse-graining.

(A3) **Deep barrier:** $\Delta U_{\text{eff}}/T_{\text{eff}} \gg 1$, so escape is exponentially rare and the Arrhenius exponent dominates the prefactor.

1.3 From one-step moments to a Fokker–Planck description

Everything in §2 follows from two conditional moments of the one-step displacement $\Delta x = x_{t+1} - x_t$. Write, at fixed position x,

$$\mu(x) \;\equiv\; \frac{\mathbb{E}[\Delta x \mid x_t = x]}{\Delta t}, \qquad D_{\text{eff}}(x) \;\equiv\; \frac{\text{Var}[\Delta x \mid x_t = x]}{2\Delta t}. \tag{3}$$

From (1), $\Delta x = v a_t \Delta t + \sqrt{2D\Delta t}\, \xi_t$ with ξ_t independent of a_t, so

$$\mu(x) = v\, \mathbb{E}[a_t \mid x], \qquad D_{\text{eff}}(x) = D + \frac{v^2 \Delta t}{2}\, \text{Var}[a_t \mid x]. \tag{4}$$

Because $a_t \in \{-1, +1\}$ we have $a_t^2 \equiv 1$, hence $\text{Var}[a_t \mid x] = 1 - \mathbb{E}[a_t \mid x]^2$, and the two moments are controlled by the *single* quantity $\mathbb{E}[a_t \mid x]$:

$$\boxed{\mu(x) = v\, \mathbb{E}[a_t \mid x], \qquad D_{\text{eff}}(x) = D + \frac{v^2 \Delta t}{2}\Big(1 - \tfrac{\mu(x)^2}{v^2}\Big).} \tag{5}$$

This is worth internalizing: for a binary-action agent, drift and diffusivity are not independent knobs. Strong drift necessarily means weak action noise. All of §2 reduces to computing $\mathbb{E}[a_t \mid x]$ for each policy class.

Remark 1. *Equation (5) uses only* one-step *statistics. If the action sequence $\{a_t\}$ is temporally correlated, the long-time diffusivity is* not *given by (5); it acquires a Green–Kubo correction $D_{\text{eff}}^{\infty} = \frac{v^2\Delta t}{2}\big[1 + 2\sum_{k\geq 1}\langle a_t a_{t+k}\rangle\big] + D$. For the 0-bit agent the actions are independent by construction and the correction vanishes. For the 1-bit agent it does* not *necessarily vanish — see §4, where it turns out to matter a great deal.*

2 Marginalized drift and diffusivity for hand-coded policies (1.1)

2.1 1.1.1 — The 0-bit case, formalized

A memoryless policy is a map $\pi : \tilde{c} \mapsto a$. Conditioning on position x and averaging over the sensing noise only:

$$\mathbb{E}[a_t \mid x_t = x] = \mathbb{E}_\eta\big[\pi\big(c(x) + \sigma\eta\big)\big] = \int_{-\infty}^{\infty} \pi\big(c(x) + \sigma u\big)\,\varphi(u)\,du, \tag{6}$$

with φ the standard normal density. The right-hand side depends on x *only* through the number $c(x)$: the integration variable u has been integrated out at fixed $c(x)$, and no other x-dependence survives. Defining

$$F(c) \;\equiv\; \int \pi(c + \sigma u)\,\varphi(u)\,du, \qquad G(c) \;\equiv\; \tfrac{1}{2}\big(1 - F(c)^2\big), \tag{7}$$

we obtain the two structural statements the Week 2 numerics must reproduce:

Key result

$$\mu(x) = v\,F\big(c(x)\big), \qquad D_{\text{eff}}(x) = D + v^2\Delta t\,G\big(c(x)\big). \tag{8}$$

Both are functions of the *concentration value* alone — not of position, not of gradient.

Worked example (threshold policy). For the natural choice $\pi(\tilde{c}) = +1$ if $\tilde{c} > \theta$ and -1 otherwise,

$$F(c) = \Pr(c + \sigma\eta > \theta) - \Pr(c + \sigma\eta \leq \theta) = 2\Phi\Big(\frac{c - \theta}{\sigma}\Big) - 1,$$

so $\mu(x) = v\big[2\Phi\big((c(x) - \theta)/\sigma\big) - 1\big]$, manifestly a function of $c(x)$ alone.

Consequence: the no-confinement proposition.

Proposition 1 (No memoryless confinement)**.** *Let x^* be an interior maximum of c, and let $x_L < x^* < x_R$ satisfy $c(x_L) = c(x_R)$. Then for every memoryless policy π and every σ,*

$$\mu(x_L) = vF\big(c(x_L)\big) = vF\big(c(x_R)\big) = \mu(x_R).$$

Confinement requires $\mu(x_L) > 0 > \mu(x_R)$, which is impossible for two equal numbers unless both vanish. No memoryless policy generates a confining drift about any peak.

Accumulation near a peak is still possible for a 0-bit agent, but only *orthokinetically* — through the spatial dependence of D_{eff} in (8), i.e. the agent lingers where it moves slowly. This is a weaker, non-exponential mechanism, and its magnitude depends on the coarse-graining convention discussed in §5. That is why the 0-bit escape is diffusive rather than Arrhenius.

2.2 1.1.2 — The 1-bit temporal-comparison policy: closed form

The policy. The agent stores one bit, the current direction of travel, and updates it by a temporal comparison:

$$a_t = \begin{cases} +a_{t-1}, & \Delta\tilde{c}_t > 0 \quad \text{(persist)}, \\ -a_{t-1}, & \Delta\tilde{c}_t \le 0 \quad \text{(reverse)}, \end{cases} \qquad \Delta\tilde{c}_t \equiv \tilde{c}_t - \tilde{c}_{t-1}. \tag{9}$$

Step 1: the statistics of the observed change. Over the step that just ended, the agent moved with action $a \equiv a_{t-1}$, so by (A1)

$$\Delta c_t = c(x_t) - c(x_{t-1}) \simeq c'(x)\big[va\Delta t + \sqrt{2D\Delta t}\,\xi\big]. \tag{10}$$

The observed change adds two *independent* sensing errors:

$$\Delta\tilde{c}_t = \Delta c_t + \sigma(\eta_t - \eta_{t-1}), \qquad \mathrm{Var}\big[\sigma(\eta_t - \eta_{t-1})\big] = 2\sigma^2. \tag{11}$$

Hence

$$\Delta\tilde{c}_t \;\sim\; \mathcal{N}\Big(\, c'(x)\, v\, a\, \Delta t\;,\;\; 2\sigma^2 + 2D\Delta t\, c'(x)^2 \,\Big). \tag{12}$$

Step 2: the decision SNR. Define

$$\boxed{s(x) \;\equiv\; \frac{|c'(x)|\, v\Delta t}{\sqrt{2\sigma^2 + 2D\Delta t\, c'(x)^2}} \;\xrightarrow[\text{sensing-dominated}]{}\; \frac{|c'(x)|\, v\Delta t}{\sqrt{2}\,\sigma}.} \tag{13}$$

The second form is the one quoted in the master document. The two contributions to the decision-noise budget are in the ratio

$$R \;=\; \frac{D\Delta t\, c'^2}{\sigma^2} \;=\; \frac{s_0^2}{\mathrm{Pe}_c}, \qquad s_0 \equiv \frac{|c'|v\Delta t}{\sqrt{2}\sigma}, \quad \mathrm{Pe}_c \equiv \frac{v^2\Delta t}{2D}, \tag{14}$$

so in the weak-signal regime ($s_0 \ll 1$) the motility contribution to the *decision* is negligible unless $\mathrm{Pe}_c \lesssim s_0^2$, which is a far more extreme condition than $\mathrm{Pe}_c \lesssim 1$. Motility noise still enters T_{eff} through D in (5); it simply does not corrupt the gradient *reading*. Keeping these two roles of D separate avoids a common confusion.

Step 3: the two-state memory chain. Let $U_t = 1$ if the agent is currently moving up-gradient, i.e. $a_t\,\mathrm{sign}(c'(x)) = +1$, and $U_t = 0$ otherwise. From (12):

- *Currently up* ($a\,\mathrm{sign}(c') = +1$): the mean of $\Delta\tilde{c}$ is $+|c'|v\Delta t$, so $\Pr(\text{persist}) = \Pr(\Delta\tilde{c} > 0) = \Phi(s)$, and persisting keeps it up.
- *Currently down* ($a\,\mathrm{sign}(c') = -1$): the mean is $-|c'|v\Delta t$, so $\Pr(\text{persist}) = \Phi(-s) = 1 - \Phi(s)$ and $\Pr(\text{reverse}) = \Phi(s)$; reversing sends it up.

The transition matrix on $\{\text{up}, \text{down}\}$ is therefore

$$P \;=\; \begin{pmatrix} \Phi(s) & 1-\Phi(s) \\ \Phi(s) & 1-\Phi(s) \end{pmatrix}, \tag{15}$$

whose two rows are identical: the probability of *arriving* in the up state is $\Phi(s)$ regardless of where the agent came from. The chain forgets its state in a single step, and the stationary occupancy is read off immediately:

$$P_\infty(\text{up}) = \Phi(s), \qquad P_\infty(\text{down}) = 1 - \Phi(s). \tag{16}$$

The coincidence is not an accident: staying up requires reading a rise correctly, and turning around from down requires reading a fall correctly, and both have probability $\Phi(s)$ because the two hypotheses are symmetric about zero.

Step 4: drift. Using (16),

$$\mathbb{E}[a_t \mid x] = \operatorname{sign}(c')\big[\Phi(s) - (1-\Phi(s))\big] = \operatorname{sign}(c')\big[2\Phi(s) - 1\big]. \tag{17}$$

Now use the identity $2\Phi(s) - 1 = \operatorname{erf}(s/\sqrt{2})$ together with $s/\sqrt{2} = |c'|v\Delta t/(2\sigma)$ from (13):

Key result

$$\mu(x) \;=\; v\,\operatorname{sign}\big(c'(x)\big)\,\operatorname{erf}\!\left(\frac{|c'(x)|\,v\Delta t}{2\sigma}\right). \tag{18}$$

Step 5: diffusivity. From (5), $\operatorname{Var}[a] = 1 - (2\Phi(s)-1)^2 = 4\Phi(s)\big(1-\Phi(s)\big)$, hence

Key result

$$D_{\text{eff}}(x) \;=\; D + 2v^2\Delta t\,\Phi(s)\big(1-\Phi(s)\big). \tag{19}$$

Equations (18)–(19) reproduce Eq. (9) of the analytical-foundations document exactly.

2.3 1.1.3 — Strong-signal limit ($s \gg 1$)

As $s \to \infty$, $\Phi(s) \to 1$ and $\operatorname{erf}(s/\sqrt{2}) \to 1$, so

$$\mu \;\longrightarrow\; v\,\operatorname{sign}(c'), \qquad D_{\text{eff}} \;\longrightarrow\; D. \tag{20}$$

The agent essentially never misreads the gradient: it climbs ballistically at full speed, and the only residual randomness is the environmental diffusion D. The approach is Gaussian-fast, since $\Phi(s)(1-\Phi(s)) \sim \varphi(s)/s \sim e^{-s^2/2}$; action noise is switched off exponentially in s.

Warning

This regime is analytically *unattractive* for the project, for two reasons. First, $D_{\text{eff}} \to D$ means the escape channel is purely thermal and carries no signature of sensing. Second, μ saturates at $\pm v$ and loses all dependence on $|c'|$, so the drift no longer derives from a potential proportional to c and the effective barrier is no longer proportional to ΔU_{trap}. The clean scaling law of §3 does not exist here.

2.4 1.1.4 — Weak-signal limit ($s \ll 1$)

Expand $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}}\big(z - z^3/3 + \dots\big)$ with $z = |c'|v\Delta t/(2\sigma)$:

$$\mu(x) \simeq v\,\operatorname{sign}(c') \cdot \frac{2}{\sqrt{\pi}} \cdot \frac{|c'|v\Delta t}{2\sigma} = \frac{v^2\Delta t}{\sqrt{\pi}\,\sigma}\,c'(x), \tag{21}$$

using $\operatorname{sign}(c')|c'| = c'$. Similarly $\Phi(s) \to \frac{1}{2} + s/\sqrt{2\pi}$, so $\Phi(1-\Phi) \to \frac{1}{4} + O(s^2)$ and

$$D_{\text{eff}} \;\longrightarrow\; D + \tfrac{1}{2}v^2\Delta t. \tag{22}$$

Three properties make this the right regime to work in, and they should be stated explicitly in the write-up because each one buys something specific:

1. **Linear response.** $\mu \propto c'(x)$, so the drift derives from a potential proportional to $-c(x)$. The effective barrier is then *proportional* to the concentration barrier ΔU_{trap}, which is what makes a clean Arrhenius slope exist at all.

2. **Sensing-limited mobility.** The proportionality constant scales as $1/\sigma$. This is the signature that performance is limited by information, not by propulsion.

3. **Additive noise.** D_{eff} is position-independent to leading order. The convention ambiguity of §5 therefore *disappears*, and the effective potential is unambiguous. This is not a minor convenience: it is the reason the prediction of §3 is parameter-free rather than convention-dependent.

3 The weak-signal effective potential and the parameter-free escape law (1.2)

3.1 1.2.1 — Constructing U_{eff} and T_{eff}

In the weak-signal regime D_{eff} is constant, so the process is an ordinary additive-noise diffusion $dx = \mu\, dt + \sqrt{2D_{\text{eff}}}\, dW$ with $\mu = -U'_{\text{eff}}$. Integrating (21):

Key result

$$U_{\text{eff}}(x) = -\frac{v^2 \Delta t}{\sqrt{\pi}\,\sigma}\, c(x), \qquad T_{\text{eff}} = D_{\text{eff}} = D + \tfrac{1}{2} v^2 \Delta t. \tag{23}$$

The stationary density is Boltzmann-like, $\rho_s \propto e^{-U_{\text{eff}}/T_{\text{eff}}}$, confirming that T_{eff} plays the role of temperature. Note the signs: at x_1 (a maximum of c) we have $c'' < 0$, so $U''_{\text{eff}}(x_1) > 0$ — a potential *well*, as required; at $x_{\min}$ (a minimum of c) $U''_{\text{eff}} < 0$ — a potential *barrier*. The picture is consistent.

Dimensional check: $[\mu] = \text{length/time}$, so $[U_{\text{eff}}] = [\int \mu\, dx] = \text{length}^2/\text{time} = [D_{\text{eff}}]$, and $U_{\text{eff}}/T_{\text{eff}}$ is dimensionless as required.

3.2 1.2.2 — The headline prediction

The effective barrier is $\Delta U_{\text{eff}} = U_{\text{eff}}(x_{\min}) - U_{\text{eff}}(x_1) = \frac{v^2\Delta t}{\sqrt{\pi}\sigma}\big[c(x_1) - c(x_{\min})\big] = \frac{v^2\Delta t}{\sqrt{\pi}\sigma}\Delta U_{\text{trap}}$. Therefore

$$\frac{\Delta U_{\text{eff}}}{T_{\text{eff}}} = \frac{v^2 \Delta t\, \Delta U_{\text{trap}}}{\sqrt{\pi}\,\sigma\,\big(D + \frac{1}{2} v^2 \Delta t\big)} \xrightarrow[D \to 0]{} \frac{v^2 \Delta t\, \Delta U_{\text{trap}}}{\sqrt{\pi}\,\sigma \cdot \frac{1}{2} v^2 \Delta t} = \frac{2}{\sqrt{\pi}}\, \frac{\Delta U_{\text{trap}}}{\sigma}. \tag{24}$$

Every dimensional parameter — v, Δt, D — has cancelled.

Key result

$$\langle T_{\text{esc}} \rangle \;\sim\; \tau_0 \exp\!\left(\frac{2}{\sqrt{\pi}}\, \frac{\Delta U_{\text{trap}}}{\sigma} \right), \qquad \frac{2}{\sqrt{\pi}} = 1.128379\ldots \tag{25}$$

The barrier is measured in units of the *sensing noise*, and the coefficient is a pure number.

3.3 1.2.3 — What this prediction does and does not assume

The slope $2/\sqrt{\pi}$ is parameter-free only inside a specific window. Each condition below is a way the test can fail for reasons that have nothing to do with memory, and each should be checked before the slope is reported.

(V1) **Weak signal along the escape path:** $s(x) \ll 1$ for all x between x_1 and $x_{\min}$ — in particular where $|c'|$ is largest, which is on the flank, *not* at the endpoints where $c' = 0$. The binding constraint is $\max_{[x_1, x_{\min}]} |c'|\, v\Delta t \ll \sqrt{2}\,\sigma$.

(V2) **Action-noise domination:** $D \ll \frac{1}{2}v^2\Delta t$, i.e. $\mathrm{Pe}_c \gg 1$. At finite D the slope is reduced by the factor $\frac{1}{2}v^2\Delta t/(D + \frac{1}{2}v^2\Delta t)$, which is a *predictable* correction, not a failure — and the crossover at $\mathrm{Pe}_c \sim 1$ is a legitimate second deliverable.

(V3) **Deep barrier:** $\Delta U_{\mathrm{eff}}/T_{\mathrm{eff}} \gg 1$, otherwise the prefactor is not negligible and the local slope is not the asymptotic slope (see §6).

(V4) **Smoothness:** $v\Delta t \ll w_1$, assumption (A1), without which the one-step expansion of Δc is invalid.

Warning

The prefactor is not parameter-free, and it depends on σ. From the Kramers form $\tau_0 = 2\pi/\sqrt{U''_{\mathrm{eff}}(x_1)|U''_{\mathrm{eff}}(x_{\min})|}$ and (23),

$$\tau_0 = \frac{2\pi\sqrt{\pi}\,\sigma}{v^2\Delta t\,\sqrt{|c''(x_1)|\,c''(x_{\min})}} \quad \propto \quad \sigma.$$

Hence $\ln\langle T_{\mathrm{esc}}\rangle = \ln\sigma + \mathrm{const} + 1.128\,\Delta U_{\mathrm{trap}}/\sigma$. *Sweep* ΔU_{trap} *at fixed* σ, where τ_0 is constant and the line is clean. A sweep over σ at fixed ΔU_{trap} carries an extra $\ln\sigma$ and will not give a straight line — and a student who fits it anyway will report a wrong slope and blame the physics.

4 An audit of the independence assumption — and why it changes the target numbers

4.1 Where the assumption entered

Step 3 of §2.2 treated the memory chain as if each decision used a *fresh* pair of readings. But the policy (9) as specified — and as specified in the master document — computes $\Delta\tilde{c}_t = \tilde{c}_t - \tilde{c}_{t-1}$ from a *single sensor read per step*. Consecutive decisions therefore share a reading:

$$\Delta\tilde{c}_t = \Delta c_t + \sigma(\eta_t - \eta_{t-1}), \qquad \Delta\tilde{c}_{t+1} = \Delta c_{t+1} + \sigma(\eta_{t+1} - \eta_t), \tag{26}$$

so $\mathrm{Cov}(\Delta\tilde{c}_t, \Delta\tilde{c}_{t+1}) = -\sigma^2 \neq 0$. Successive comparisons are *anti-correlated*, the two rows of P are no longer the whole story, and (16) is not exact. The correct state is the pair (a_t, η_t), not a_t alone.

4.2 The exact stationary condition

Let $h(u) = \Pr(a = +1 \mid \eta = u)$ in stationarity, take $\mathrm{sign}(c') = +1$, and write $\varepsilon = |c'|v\Delta t/\sigma$. Since η_{t+1} is drawn independently, conditioning on the new noise value gives the linear integral equation

$$h(u') \;=\; H(u'+\varepsilon) + H(u'-\varepsilon) + 1 - \Phi(u'-\varepsilon) - H(\infty), \qquad H(y) \equiv \int_{-\infty}^{y} \varphi(u)h(u)\,du, \tag{27}$$

and then $\mathbb{E}[a] = \int \varphi(u)\big[2h(u) - 1\big]du$. This has no obvious closed-form solution, but it is trivial to solve numerically on a grid.

4.3 What the numbers actually are

Numerical check

Both routes below — direct Monte Carlo of (9) and numerical solution of (27) — agree to four digits.

	independent readings	shared reading (single sensor)
μ vs. closed form (18)	1.0000	$\kappa_\mu = 0.7208$
D_{eff}^{∞} vs. one-step form (19)	1.0000	$\kappa_D = 0.523$

Both ratios are measured in the weak-signal limit $s \to 0$ and are constants there (checked over $\varepsilon \in [0.0125, 0.4]$). The independent-readings design reproduces (18)–(19) to 4–5 decimal places, confirming the algebra of §2.2 is correct *given its assumption*.

The diffusivity deficit has a clean origin: it is exactly the Green–Kubo correction of Remark 1. The shared reading makes the agent *oscillate*. Measuring the action autocorrelation on a flat landscape gives $\langle a_t a_{t+1}\rangle = 0$, $\langle a_t a_{t+2}\rangle = -1/3$, $\langle a_t a_{t+4}\rangle \approx +2/15$, with all odd lags vanishing by symmetry. The lag-2 value is exact and follows from the Gaussian arcsine law: consecutive differences of i.i.d. normals have correlation $-\frac{1}{2}$, and $\mathbb{E}[\operatorname{sign} X \operatorname{sign} Y] = \frac{2}{\pi}\arcsin\rho$, so

$$\langle a_t a_{t+2}\rangle = \frac{2}{\pi}\arcsin\!\left(-\tfrac{1}{2}\right) = -\frac{1}{3}. \tag{28}$$

Summing the series gives $D_{\text{eff}}^{\infty}/\left(\frac{1}{2}v^2\Delta t\right) = 1 + 2\sum_{k\geq 1}\langle a_t a_{t+k}\rangle \approx 0.523$.

4.4 Consequence for the headline slope

The exponent (24) is the ratio of the drift coefficient to T_{eff}, so the two corrections do *not* cancel — they compound in opposite directions:

$$\text{slope}_{\text{shared}} = \frac{\kappa_\mu}{\kappa_D}\cdot\frac{2}{\sqrt{\pi}} \approx \frac{0.7208}{0.523}\times 1.1284 \approx 1.55. \tag{29}$$

Warning

The number to test against depends on a design choice that the master document does not flag. With the single-sensor policy as literally specified, the Week 2 gate "1-bit drift matches $v\,\mathrm{erf}(\cdot)$ to a few percent" *cannot pass* — it will fail by 28%, and the Week 3 slope will come out near 1.55, not 1.128. A student who trusts the stated targets will conclude their simulator is broken when it is correct.

Two clean remedies, and a recommendation.

(a) *Change the agent* so each decision uses an independent gradient estimate (two fresh reads per step, or an explicitly resampled comparison). The closed forms (18)–(19) then become *exact*, the $2/\sqrt{\pi}$ prediction stands, and the analytics stay parameter-free. Cost: one extra sensor read per step, and slightly less biological realism.

(b) *Keep the single-sensor agent* and carry κ_μ, κ_D as measured constants. More faithful to a real bacterium, which has one receptor array; but the headline prediction is no longer parameter-free, since κ_μ and κ_D come from numerics rather than algebra.

Recommendation: adopt (a) for the Week 2–3 validation chain, because the whole point of that chain is to test code against an exact known answer. Then run (b) once as a deliberate comparison. The gap between them — an agent that reuses its last measurement performs measurably worse than one that takes a fresh one — is a genuine, reportable result about the cost of correlated sensing, and it costs almost nothing to obtain once the pipeline exists.

5 Which stochastic-calculus convention the discrete update realizes (1.3)

5.1 1.3.1 — The three candidates

When D_{eff} depends on position, the SDE $dx = \mu\,dt + \sqrt{2D_{\text{eff}}(x)}\,dW$ is ambiguous until an interpretation is fixed. The three standard choices give three different zero-flux stationary densities:

Convention	Zero-flux condition ($\mu = 0$)	ρ_s	α
Itô (pre-point)	$\partial_x(D_{\text{eff}}\rho) = 0$	$\rho \propto D_{\text{eff}}^{-1}$	1
Stratonovich (mid-point)	$\sqrt{D_{\text{eff}}}\,\partial_x(\sqrt{D_{\text{eff}}}\rho) = 0$	$\rho \propto D_{\text{eff}}^{-1/2}$	$\frac{1}{2}$
Hänggi (anti-Itô, post-point)	$D_{\text{eff}}\,\partial_x\rho = 0$	$\rho = \text{const}$	0

Including the drift, all three are captured by a single one-parameter family:

$$\rho_s(x) \;\propto\; D_{\text{eff}}(x)^{-\alpha}\,\exp\!\left(\int^x \frac{\mu(z)}{D_{\text{eff}}(z)}\,dz\right), \qquad \alpha \in \{1, \tfrac{1}{2}, 0\}. \tag{30}$$

5.2 1.3.2 — Why it must be measured, and how to measure it correctly

The two most natural readings of our own model land in *different* conventions. The update (1) draws its action from a reading taken at x_t and applies it across the whole step — a pre-point rule, hence Itô, $\alpha = 1$. But the continuous-time run-and-tumble particle with position-dependent speed satisfies $v\rho = \text{const}$ exactly, i.e. $\rho \propto D_{\text{eff}}^{-1/2}$, Stratonovich, $\alpha = \frac{1}{2}$. There is no way to settle this by staring at the equations; it must be measured.

Warning

Do not fit $\rho_s \propto D_{\text{eff}}^{-\alpha}$ directly. That naive fit is only valid when $\mu \equiv 0$. For the 0-bit agent $\mu \neq 0$, and (30) shows the drift contributes a factor that is *not* a power of D_{eff} — the exponential involves an integral over x, which is not a function of $c(x)$ alone. Fitting the naive power law will return a meaningless effective exponent somewhere between the candidates and will look deceptively convincing.

Estimator 1 (drift-corrected, works in general). Measure $\mu(x)$ and $D_{\text{eff}}(x)$ from the same rollouts, form the drift-subtracted log-density

$$\Psi(x) \;\equiv\; \ln \rho_s(x) \;-\; \int^{x} \frac{\mu(z)}{D_{\text{eff}}(z)}\, dz, \tag{31}$$

and regress Ψ on $\ln D_{\text{eff}}$. By (30) the slope is exactly $-\alpha$.

Estimator 2 (drift-free, cleanest). Build a diagnostic agent for which $\mu \equiv 0$ but D_{eff} still varies. With binary actions this is impossible, since $\mathbb{E}[a] = 0$ forces $\mathrm{Var}[a] = 1$. Allowing $a \in \{-1, 0, +1\}$ makes it easy: choose $\Pr(\pm 1) = p(c)$ and $\Pr(0) = 1 - 2p(c)$, giving $\mu \equiv 0$ and $D_{\text{eff}} = D + v^2 \Delta t\, p(c(x))$. Then the naive power-law fit *is* valid and α is read off directly. This costs a few lines of code and removes all ambiguity.

Why this protects everything downstream. Whatever α comes back, it enters the escape calculation, because the MFPT quadrature of §6 is written for Itô. Matching stationary densities between (30) and the Itô form gives the required translation:

$$\boxed{\mu_{\text{Itô}}(x) = \mu(x) + (1 - \alpha)\, D'_{\text{eff}}(x).} \tag{32}$$

An unmeasured α is therefore an unquantified error in the effective drift, hence in the barrier, hence in the fitted Arrhenius slope — which would be silently misattributed to memory. Note that in the weak-signal regime $D'_{\text{eff}} = 0$ and the correction vanishes identically: one more reason to do the headline measurement there.

6 Exact 1D mean-first-passage-time quadrature (1.4)

6.1 1.4.1 — Derivation

For Itô dynamics $dx = \mu\, dt + \sqrt{2D_{\text{eff}}}\, dW$ on $[r, b]$ with a reflecting boundary at r and an absorbing boundary at b, the MFPT $T(x)$ satisfies the backward equation

$$\mu(x) T'(x) + D_{\text{eff}}(x) T''(x) = -1, \qquad T'(r) = 0, \quad T(b) = 0. \tag{33}$$

Step 1. Introduce the integrating factor $\psi(y) = \exp\!\big(\int_r^y \mu / D_{\text{eff}}\, dz\big)$, so that $\psi' = \psi\, \mu / D_{\text{eff}}$ and $\psi(r) = 1$. Multiplying the equation by ψ / D_{eff} collapses the left side to a total derivative:

$$\big(\psi T'\big)' = -\frac{\psi}{D_{\text{eff}}}. \tag{34}$$

Step 2. Integrate from r to y and impose $T'(r) = 0$:

$$T'(y) = -\frac{1}{\psi(y)} \int_r^y \frac{\psi(z)}{D_{\text{eff}}(z)}\, dz. \tag{35}$$

Step 3. Integrate from x to b and impose $T(b) = 0$:

Key result

$$T(x) \;=\; \int_x^b \frac{dy}{\psi(y)} \int_r^y \frac{\psi(z)}{D_{\text{eff}}(z)}\, dz, \qquad \psi(y) = \exp\!\left(\int_r^y \frac{\mu(z)}{D_{\text{eff}}(z)} dz \right). \tag{36}$$

Exact for *all* barrier heights, and it handles position-dependent D_{eff} automatically.

6.2 1.4.2 — Specializing, and implementing it stably

Insert μ and D_{eff} from (18)–(19) (with the correction (32) if $\alpha \neq 1$), set r well inside the trap basin on the far side of x_1, and $b = x_{\min} + \delta$. Evaluate on the same grid as the Week 1 landscape module.

Warning

Never evaluate ψ directly. With $\Delta U_{\text{eff}}/T_{\text{eff}} \sim 6$–$10$, ψ spans many orders of magnitude and the outer/inner factors overflow and underflow separately even though their product is $O(1)$. Write $\psi = e^{-\Phi}$ with $\Phi(y) = -\int_r^y \mu/D_{\text{eff}}$, so that (36) becomes

$$T(x) = \int_x^b dy \int_r^y \frac{e^{\Phi(y)-\Phi(z)}}{D_{\text{eff}}(z)}\, dz,$$

and the exponent is bounded by the barrier height. Accumulate Φ once by cumulative trapezoid, then form the difference inside the loop. This is the difference between a routine that works to $\Delta U_{\text{trap}}/\sigma \sim 20$ and one that returns `inf` above ~ 5.

6.3 1.4.3 — Extracting a local Arrhenius slope

The asymptotic Kramers formula

$$T_{\text{saddle}} = \frac{2\pi}{\sqrt{U''_{\text{eff}}(x_1)\,|U''_{\text{eff}}(x_{\min})|}}\, \exp\!\left(\frac{\Delta U_{\text{eff}}}{T_{\text{eff}}}\right) \tag{37}$$

is a *leading-order* result. At the barrier heights actually reachable in simulation, $\Delta U_{\text{eff}}/T_{\text{eff}} \sim 2$–$6$, its prefactor is in error by tens of percent — and that error will be misread as a memory effect if it is not controlled.

Key result

Evaluate (36) on a ladder of ΔU_{trap} at fixed σ, then differentiate numerically:

$$\text{slope}_{\text{local}}(\Delta U_{\text{trap}}) \;=\; \frac{\partial \ln T}{\partial(\Delta U_{\text{trap}}/\sigma)}. \tag{38}$$

Compare the *measured* slope to this *local* slope at matched ΔU_{trap} — never to the asymptotic 1.128 unless (V3) has been verified. The local slope approaches $2/\sqrt{\pi}$ from below as the barrier deepens; plotting that approach is itself a good figure, because it shows the reader exactly where the asymptotic regime begins.

7 Summary of results and numerical targets

Quantity	Result	Eq.
0-bit drift / diffusivity	$\mu = vF(c), \quad D_{\text{eff}} = D + v^2\Delta t\, G(c)$	(8)
Decision SNR	$s = \lvert c'\rvert v\Delta t/\sqrt{2\sigma^2 + 2D\Delta t c'^2}$	(13)
1-bit drift	$\mu = v\,\text{sign}(c')\,\text{erf}\big(\lvert c'\rvert v\Delta t/2\sigma\big)$	(18)
1-bit diffusivity	$D_{\text{eff}} = D + 2v^2\Delta t\,\Phi(s)(1-\Phi(s))$	(19)
Weak-signal drift	$\mu \simeq v^2\Delta t\, c'/(\sqrt{\pi}\sigma)$	(21)
Effective potential	$U_{\text{eff}} = -v^2\Delta t\, c/(\sqrt{\pi}\sigma), \quad T_{\text{eff}} = D + \frac{1}{2}v^2\Delta t$	(23)
Headline slope	$\langle T_{\text{esc}}\rangle \sim \tau_0 e^{1.128\,\Delta U_{\text{trap}}/\sigma}$ (independent readings)	(25)
single-sensor variant	slope ≈ 1.55; $\kappa_\mu = 0.721$, $\kappa_D = 0.523$	§4
Convention family	$\rho_s \propto D_{\text{eff}}^{-\alpha} e^{\int \mu/D_{\text{eff}}}$	(30)
Convention correction	$\mu_{\text{Itô}} = \mu + (1-\alpha)D'_{\text{eff}}$	(32)
MFPT quadrature	$T(x) = \int_x^b \psi^{-1}\int_r^y \psi/D_{\text{eff}}$	(36)

Revised validation gates

(G1) 0-bit drift is symmetric about each peak, $\mu(x_L) = \mu(x_R)$ whenever $c(x_L) = c(x_R)$, to within sampling error.

(G2) Convention exponent α recovered cleanly by Estimator 1 or 2 of §5, and *reported*.

(G3) 1-bit drift matches (18) to a few percent — **using the independent-readings agent**. If the single-sensor agent is used instead, the target is κ_μ times (18), and the gate must be restated.

(G4) Long-time D_{eff} agrees with the one-step formula (19); if not, measure $\langle a_t a_{t+k}\rangle$ and account for it via Green–Kubo before proceeding (Remark 1).

(G5) At small D and fixed σ, $\ln\langle T_{\text{esc}}\rangle$ versus $\Delta U_{\text{trap}}/\sigma$ is linear with slope matching the *local* quadrature slope of §6, tending to $2/\sqrt{\pi}$ as the barrier deepens.